THE LOCUS WHERE A MAP FROM A FINITE MODULE TO A FINITELY PRESENTED MODULE IS AN ISOMORPHISM IS OPEN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted spans contain no \cite, so the card carries no bibliography.

Allowed standard prerequisites (editorial).

- A finite presentation of the target module. Used at the start of the counted proof (lines 19638-19640). Lets preimages of the generators and of the relations be written down explicitly.
- Finitely many conditions spread out to a basic open. Used at the end of the counted proof (lines 19650-19660). Produces elements whose simultaneous nonvanishing defines an open subset of the isomorphism locus.

Source reference. Original lemma 05GE: sources/stacks-algebra/pinned-original-source.tex, source lines 19601–19601. Navigation pointer for \ref{lemma-map-between-finite} used inside the counted span (line 19656); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a ring. Let $\varphi : M \to N$ be a map of R-modules with M finite and N finitely presented. Then*

$$U = \{\mathfrak{p} \subset R \mid \varphi_{\mathfrak{p}} : M_{\mathfrak{p}} \to N_{\mathfrak{p}} \text{ is an isomorphism}\}$$

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is an open subset of $\mathrm{Spec}(R)$.

Proof. Let $\mathfrak{p} \in U$. Pick a presentation $N = R^{\oplus n}/\sum_{j=1,\ldots,m} Rk_j$. Denote e_i the image in N of the ith basis vector of $R^{\oplus n}$. For each $i \in \{1, \ldots, n\}$ choose an element $m_i \in M_{\mathfrak{p}}$ such that $\varphi(m_i) = f_i e_i$ for some $f_i \in R$, $f_i \notin \mathfrak{p}$. This is possible as $\varphi_{\mathfrak{p}}$ is an isomorphism. Set $f = f_1 \ldots f_n$ and let $\psi : R_f^{\oplus n} \to M_f$ be the map which maps the ith basis vector to m_i/f_i. Note that $\varphi_f \circ \psi$ is the localization at f of the given map $R^{\oplus n} \to N$. As $\varphi_{\mathfrak{p}}$ is an isomorphism we see that $\psi(k_j)$ is an element of M which maps to zero in $M_{\mathfrak{p}}$. Hence we see that there exist $g_j \in R$, $g_j \notin \mathfrak{p}$ such that $g_j\psi(k_j) = 0$. Setting $g = g_1 \ldots g_m$, we see that ψ_g factors through N_{fg} to give a map $\chi : N_{fg} \to M_{fg}$. By construction χ is a right inverse to φ_{fg}. It follows that $\chi_{\mathfrak{p}}$ is an isomorphism. By Lemma 05GE there is an $h \in R$, $h \notin \mathfrak{p}$ such that $\chi_h : N_{fgh} \to M_{fgh}$ is surjective. Hence φ_{fgh} and χ_h are mutually inverse maps, which implies that $D(fgh) \subset U$ as desired. □